

Intellectual Property

The Role of Intellectual Property on Enterprise-led Industry-University-Research (EIUR) Institution Cooperation: Case of Innovation Awards in China

Background

National Innovation System (NIS) framework: shift from government-led to enterprise-led innovation in China.

NIS, RIS, and SIS

- NIS: National Innovation System
- RIS: Regional Innovation System
- SIS: Sectoral Innovation System
- All emphasize multi-actor interactions.

Importance of Enterprises

- Enterprises are the mainstay of innovation.
- Key in activating technological resources and improving efficiency.

Role of IP Strategy

IP protection is vital for technological progress, industrial innovation, and competitiveness.

Research Motivation

Gap: prior studies focus on patents/R&D but overlook commercialization and asymmetric power dynamics in IUR cooperation.

Objectives

- Analyze IPDC policy's effect on E-IUR cooperation.
- Examine enterprise dominance.
- Investigate mediation effects of investment and enforcement.

IPDC Policy Overview

- Launched in 2011 by CNIPA.
- Goal: strengthen IP creation, management, financing, and enforcement.
- 77 cities approved by 2022.

Theoretical Framework

- IPDC enhances cooperation via:
- - Strengthening IP awareness
- - Building service platforms
- - Supporting enterprise leadership.

Research Hypotheses

- H1: IPDC strengthens E-IUR cooperation.
- H2a: Works via enterprise innovation investment.
- H2b: Works via regional IP law enforcement (IPLE).

Data & Sample

- Sample: 3,152 listed firms (2008–2020).
- Sources: WIND, CSMAR, NOSTA, CNIPA.
- ~26,496 firm-year observations.

Methodology

- Approach: Difference-in-Differences (DID) regression.
- Treatment: firms in IPDC cities.
- Outcome: SSTP Awards as measure of deep E-IUR integration.

Variables

- Dependent: E-IUR cooperation (via awards).
- Independent: IPDC policy.
- Controls: Firm age, size, ownership, leverage, ROA, etc.

Results: Main Findings

- IPDC policy significantly strengthens E-IUR cooperation (positive DID coefficient).
- Hypothesis H1 supported.

Robustness Tests

- Parallel trend satisfied.
- PSM-DID (Propensity matching score methods) confirms robustness.
- Endogenous issues controlled with two-stage treatment effects.

Mechanism Analysis

Two mediation channels confirmed:

1. Increased enterprise R&D investment (H2a).
2. Enhanced regional IP enforcement (H2b).

Further Insights

Comparison with IP Demonstration Enterprises (IPDE):

- Stronger effect than IPDC policy.
- Highlights role of enterprise-specific IP initiatives.

Discussion

Key contributions:

- Commercialization focus using awards.
- Asymmetric dependence acknowledged.
- Integration of IP + IUR framework.

Policy Implications

- Government: refine IPDC, promote replication nationwide, improve enforcement.
- Support more cities in IPDC construction.

Conclusion

IPDC policy significantly deepens E-IUR cooperation by stimulating enterprise innovation and strengthening IP law enforcement.